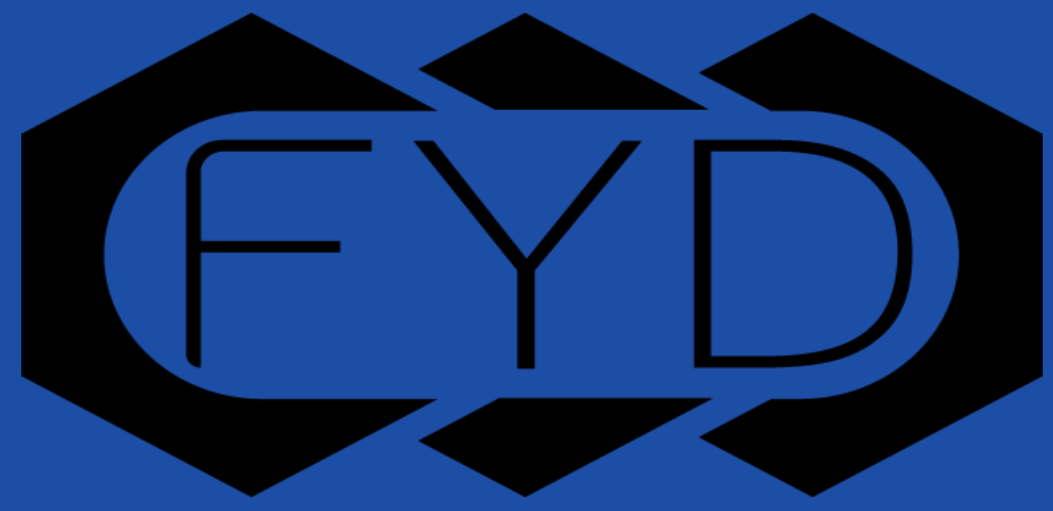
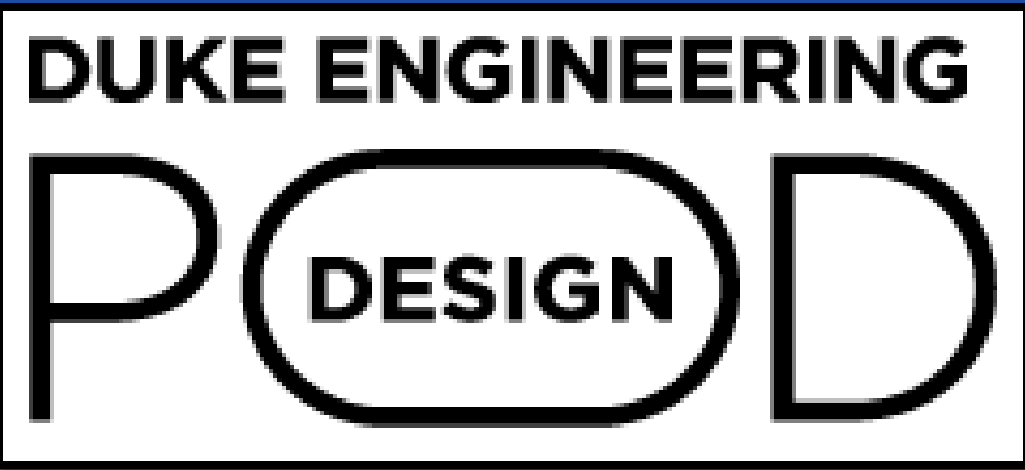




Medication Adherence Alarm for Adults With ADHD

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Design Project Goal

This project addresses the lack of a straightforward and dependable system that helps adults with ADHD maintain consistent medication routines.

In response, our design goal is to create a **reliable and accessible reminder device** to support daily medication adherence.

Background and Motivation

- Missed ADHD medication doses are frequent due to distraction, forgetfulness, and inconsistent daily routines.
- Existing reminders (e.g., alarms, pillboxes) are easily dismissed or ignored, reducing treatment effectiveness.
- Individuals with ADHD benefit from systems that **support habit formation** while **minimizing cognitive demands**.
- A **reliable, multi-sensory reminder** paired with a **simple check-in** can improve medication adherence.

Design Objectives

Criterion	Target	Result
Detectability	>90% of users notice alerts	PASS - 5/5 users detected alerts immediately
Setup Time	Device setup in < 30 seconds	PASS - 3 users: ~30 sec, 2 users: ~1 min
Connectivity	Maintain BLE connection at ≥30 m	PASS - Stable connectivity across test distance
Battery Life	≥4 days of continuous operation	FAIL - WiFi-only: 14 hrs PASS - RTC-enabled: ~14 days
NFC Integration	Reliable NFC + BLE logging	PASS - 20/20 successful NFC scans

Design Solution

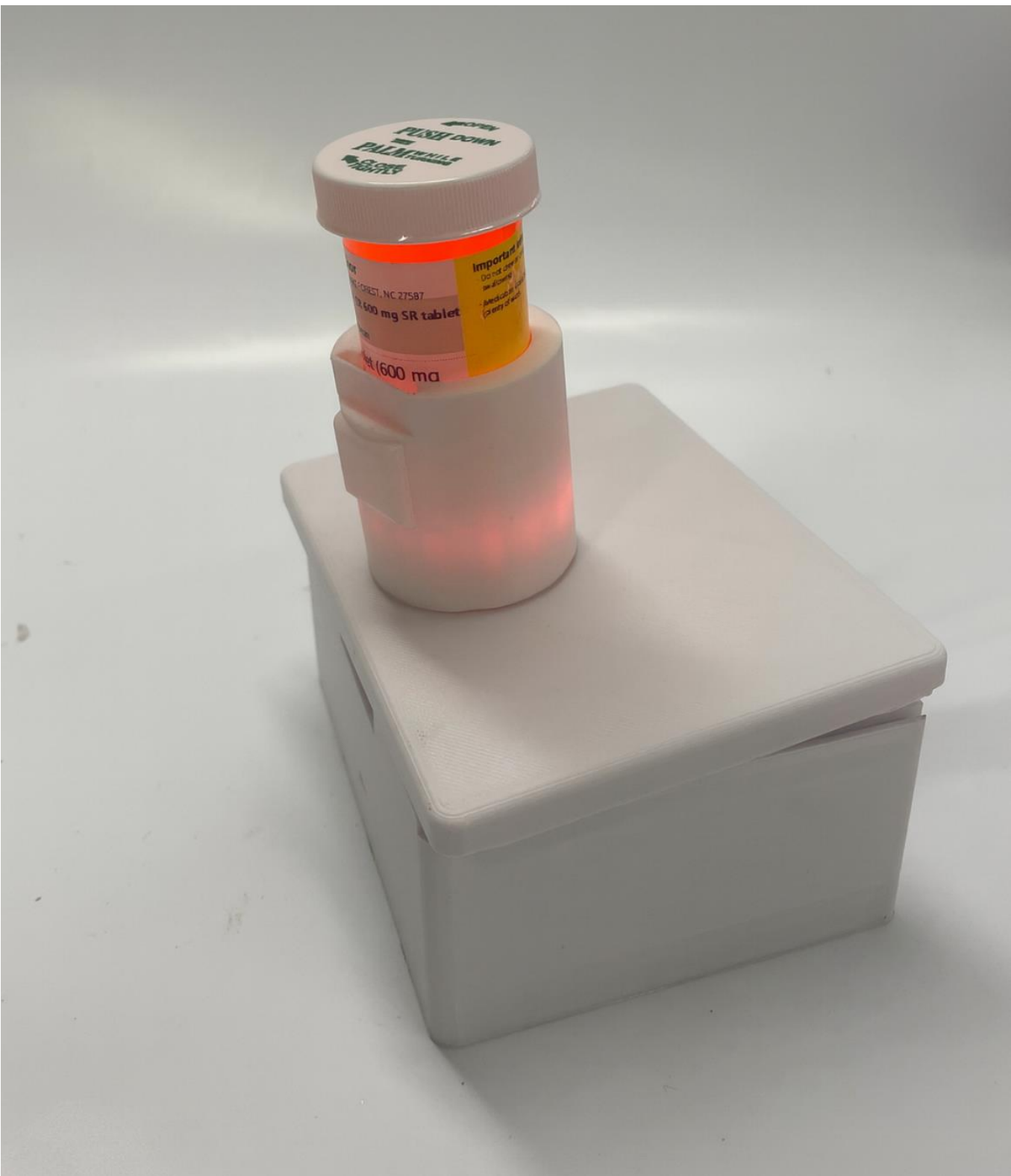


Figure 1. Pill bottle module lit red during alert.



Figure 2. Final wristband prototype with vibration motor.

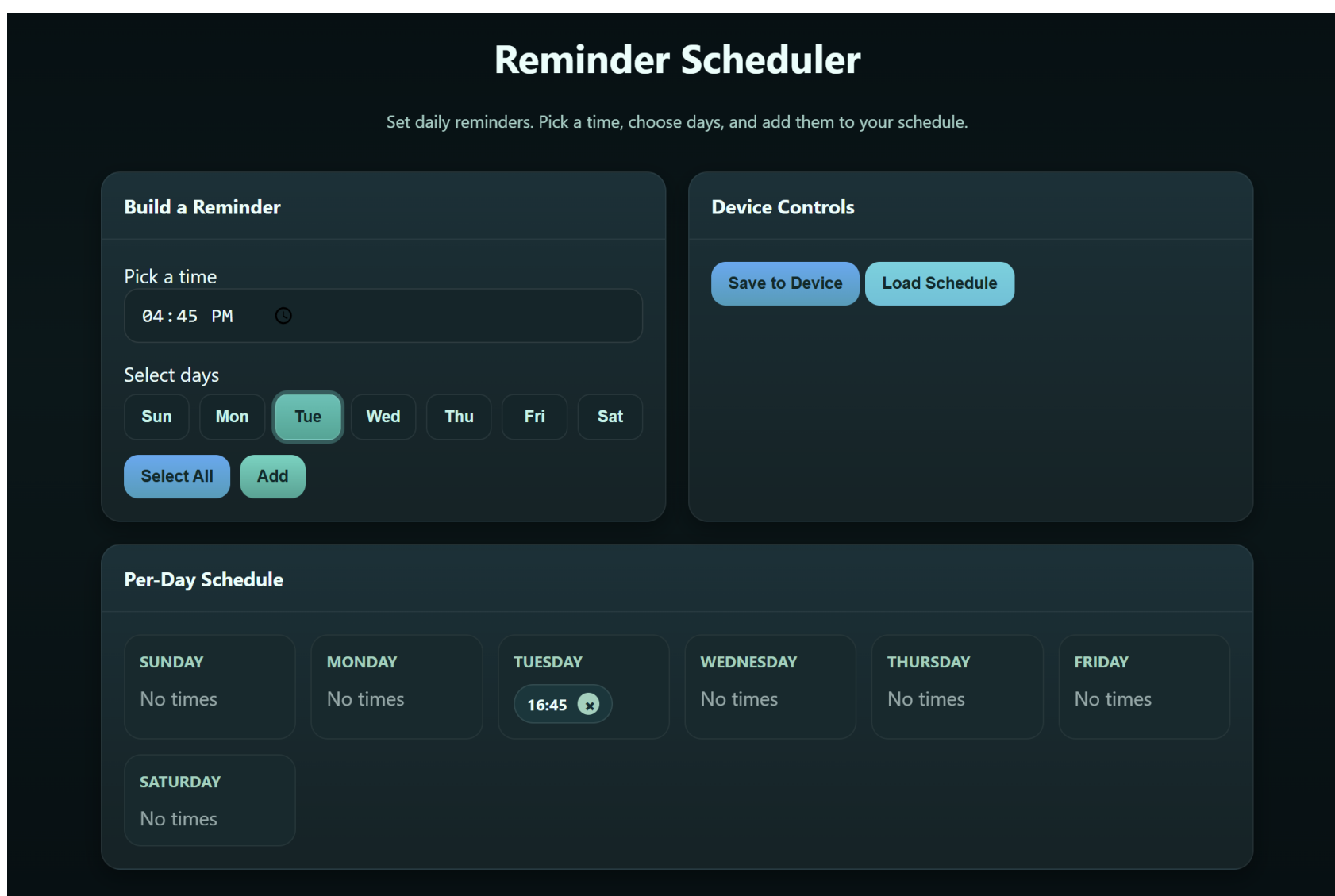
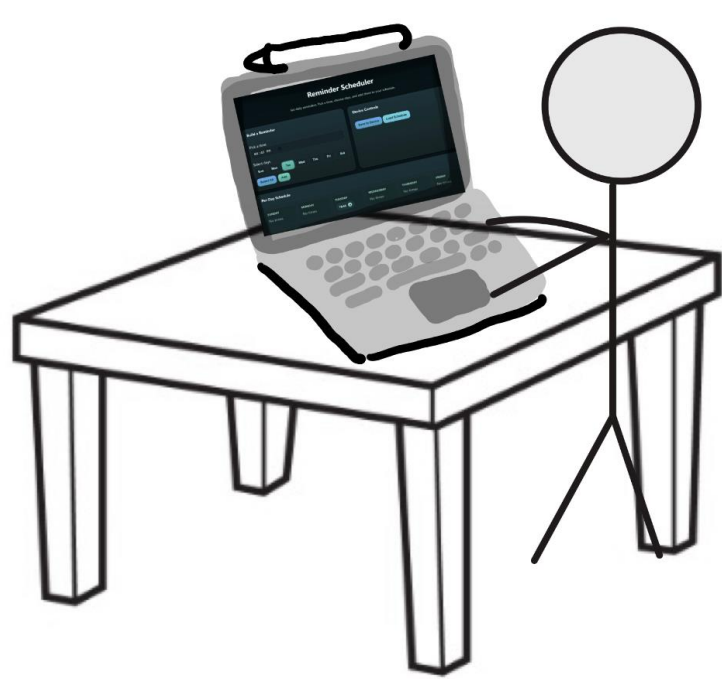
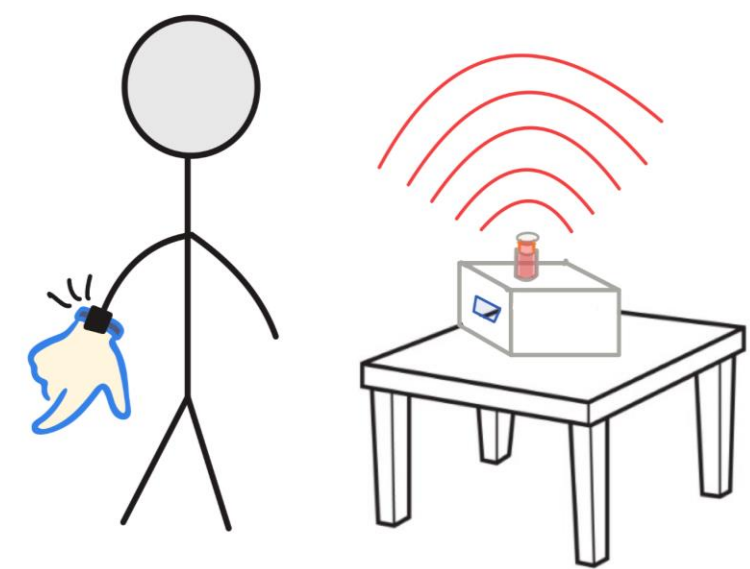


Figure 3. Web interface for scheduling and syncing reminders.

How It Works



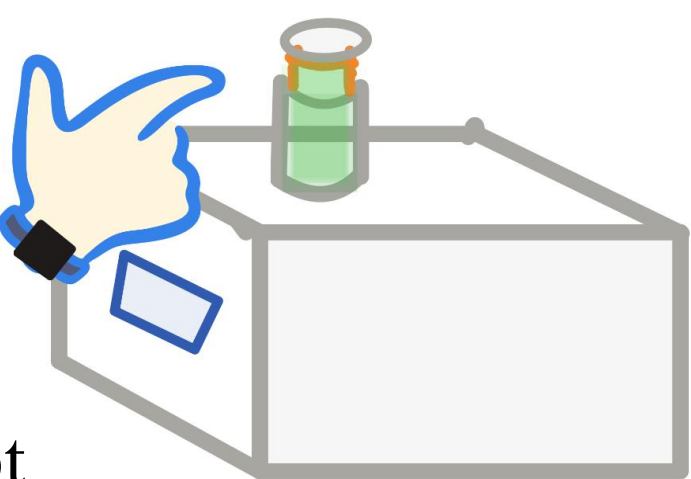
4a. User configures device using web interface to input medication schedule



4b. Device issues multisensory alert to prompt user to take medication at scheduled times

Figure 4. Workflow sketches showing scheduling, alert detection, and NFC confirmation.

4c. User deactivates alert by tapping device with NFC tag to confirm completion



Conclusion

Our prototype delivers reliable, easy-to-interpret alerts while meeting core requirements for detectability, battery life, setup time, and connectivity. These results demonstrate the system's feasibility as a practical assistive tool for adults with ADHD.

In the future, we aim to:

- Reduce wristband size to **improve wearability and user comfort**
- Transition to a custom PCB for **greater durability and system integration**
- Expand user testing to **evaluate long-term performance and real-world adherence**

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